

Lecture 5: Joint and Marginal Distributions, Conditioning, Independence

Jamie Haddock

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1 Why look at two random variables together?

So far we've studied one random variable at a time. But often we care about how *two* random variables move together — two dice rolled at once, or (in randomized Kaczmarz) the row index sampled at one iteration

versus the next. Knowing each RV's own PMF separately says nothing about how they *relate*.

i Definition: Joint PMF

For discrete random variables X, Y on the same probability space,

$$p_{X,Y}(x, y) = P(X = x, Y = y).$$

Like any PMF, it's nonnegative and sums to 1 over all pairs (x, y) . It's strictly more informative than p_X and p_Y separately — it tells us how X and Y *co-vary*.

Two-die-roll example: Let X = first die, Y = second die — all 36 outcomes $(x, y) \in \{1, \dots, 6\}^2$ are equally likely, so

$$p_{X,Y}(x, y) = \frac{1}{36}, \quad x, y \in \{1, \dots, 6\}.$$

A Joint distributions of arbitrarily many RVs!

We may consider joint PMFs of any number of RVs defined on the same probability space, $p_{\{X_i\}_{i=1}^k}(\{x_i\}_{i=1}^k) = P(X_1 = x_1, \dots, X_k = x_k)$.

2 Marginal PMF

i Definition: Marginal PMF

Given a joint PMF $p_{X,Y}$, the **marginal PMF** of X is recovered by summing out y :

$$p_X(x) = P(X = x) = \sum_y p_{X,Y}(x, y),$$

and symmetrically $p_Y(y) = \sum_x p_{X,Y}(x, y)$.

Two-die-roll example: summing each row of the 36-entry joint table over y gives

$$p_X(x) = \sum_{y=1}^6 \frac{1}{36} = \frac{6}{36} = \frac{1}{6}, \quad x = 1, \dots, 6,$$

recovering the familiar single-die PMF.

3 Expectation of a function of two RVs

i Theorem: LOTUS for two random variables

For random variables X, Y and any function g ,

$$E[g(X, Y)] = \sum_{x,y} g(x, y) p_{X,Y}(x, y).$$

Taking $g(x, y) = ax + by + c$ recovers linearity of expectation that we saw previously:

$$\begin{aligned} E[aX + bY + c] &= \sum_{x,y} (ax + by + c) p_{X,Y}(x, y) = a \sum_x x p_X(x) + b \sum_y y p_Y(y) + c \\ &= aE[X] + bE[Y] + c. \end{aligned}$$

(The middle step uses the marginal-PMF definition — summing y out of $ax p_{X,Y}(x, y)$ leaves $ax p_X(x)$, and likewise for the y term.)

4 Example: the hat-check problem

n people check their hats at a party; the hats are returned to them in a uniformly random order (a uniformly random permutation of the n hats). Let X = number of people who get their own hat back. Find $E[X]$.

Write $X = \sum_{i=1}^n X_i$, where $X_i = \mathbf{1}_{\{\text{person } i \text{ gets their own hat back}\}}$. By symmetry every person is equally likely to receive any of the n hats, so $P(X_i = 1) = \frac{1}{n}$ and $E[X_i] = \frac{1}{n}$.

⚠ The X_i are *not* independent!

If the first $n - 1$ people all get their own hat back, the last person is guaranteed to as well — knowing some X_i 's tells you a lot about the others. But, just like the expectation of the Binomial calculation we saw previously, **linearity doesn't care**:

$$E[X] = \sum_{i=1}^n E[X_i] = n \cdot \frac{1}{n} = 1.$$

However many people are at the party, on average exactly **one** person gets their own hat back!

5 Conditioning

We've used conditional probability previously — let's now define *conditional distributions* of random variables precisely.

i Definition: Conditional PMF given an event

For an event A with $P(A) > 0$,

$$p_{X|A}(x) = P(X = x \mid A) = \frac{P(\{X = x\} \cap A)}{P(A)}.$$

Revisiting our die-roll example: let X be a single fair die roll and $A = \{X \text{ is even}\} = \{2, 4, 6\}$. Then

$$p_{X|A}(x) = \begin{cases} \frac{1}{3} & x \in \{2, 4, 6\} \\ 0 & \text{otherwise,} \end{cases}.$$

6 Conditioning on another random variable

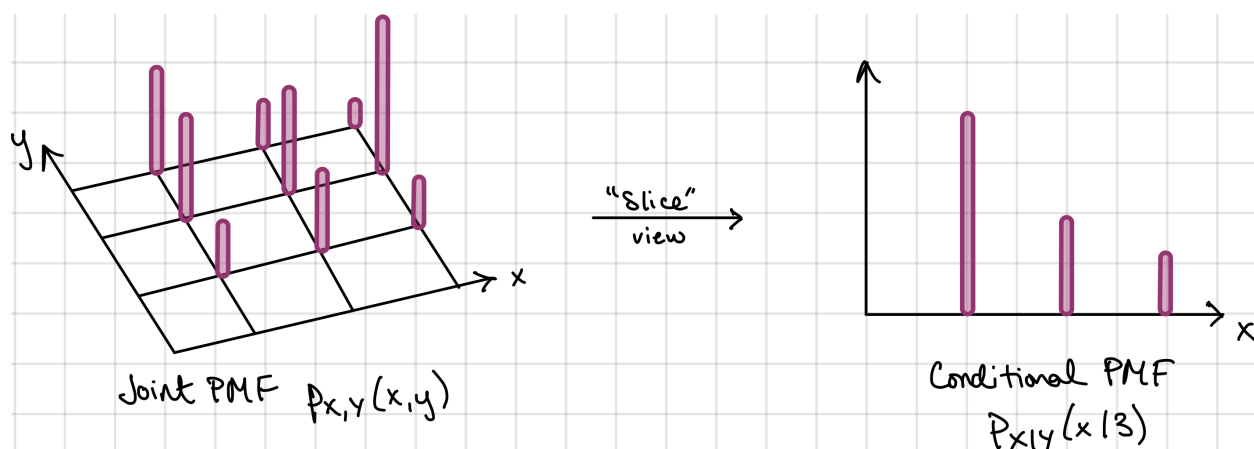
i Definition: Conditional PMF given $Y = y$

For $p_Y(y) > 0$,

$$p_{X|Y}(x | y) = P(X = x | Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

This is exactly the previous definition applied to the event $A = \{Y = y\}$.

7 Grid view of the joint PMF, slice view of the conditional PMF



8 Recovering the marginal from conditional PMFs

i Theorem (total probability, for random variables)

$$p_X(x) = \sum_y p_Y(y) p_{X|Y}(x | y).$$

This is just $p_X(x) = \sum_y p_{X,Y}(x, y)$ (the marginal definition) with $p_{X,Y}(x, y) = p_Y(y) p_{X|Y}(x|y)$ substituted in.

9 Conditional expectation

i Definitions: conditional expectation

$$E[X | A] = \sum_x x p_{X|A}(x), \quad E[X | Y = y] = \sum_x x p_{X|Y}(x | y).$$

Same idea as ordinary expectation — just averaging x against the *conditional* PMF instead of the unconditional one.

10 Law of total expectation – three forms

- **Partition form:** if $A_1, \dots, A_n$ partition the sample space,

$$E[X] = \sum_{i=1}^n P(A_i) E[X | A_i].$$

- **Conditioned partition form:** for any event B ,

$$E[X | B] = \sum_{i=1}^n P(A_i | B) E[X | A_i \cap B].$$

- **Random-variable form:**

$$E[X] = \sum_y p_Y(y) E[X | Y = y].$$

All three say the same thing: split the world into (disjoint) pieces, compute the (conditional) expectation on each piece, and average the pieces back together weighted by how likely each one is.

This is a useful computational trick: whenever a direct computation of $E[X]$ looks hard, look for a natural way to split on “what happened first,” compute the easy conditional pieces, and average.

11 Example: mean and variance of the Geometric, the slick way

Let $X \sim \text{Geometric}(p)$ (number of trials until the first success). Condition on the first trial: $A = \{\text{trial 1 succeeds}\}$, $P(A) = p$.

- If A occurs, $X = 1$, so $E[X | A] = 1$.
- If A^c occurs, that trial was wasted and — by memorylessness — we’re back to square one: $X = 1 + X'$ for a fresh $\text{Geometric}(p)$ copy X' , so $E[X | A^c] = 1 + E[X]$.

By the law of total expectation,

$$E[X] = P(A)E[X | A] + P(A^c)E[X | A^c] = p \cdot 1 + (1 - p)(1 + E[X]) = 1 + (1 - p)E[X],$$

so $p E[X] = 1$, giving $E[X] = \frac{1}{p}$ — which matches our previous calculation.

The same trick on X^2 : $X^2 | A^c = (1 + X')^2 = 1 + 2X' + X'^2$, so

$$E[X^2] = p + (1 - p)(1 + 2E[X] + E[X^2]) = 1 + 2(1 - p)E[X] + (1 - p)E[X^2],$$

which solves to $E[X^2] = \frac{2 - p}{p^2}$, and therefore $\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{1 - p}{p^2}$ — which is what we saw in a previous lecture!

12 Recursing the randomized Kaczmarz bound

Recall our previous one-step bound, valid conditional on the *current* iterate x_k :

$$E[\|x_{k+1} - x^\star\|^2 | x_k] \leq (1 - c) \|x_k - x^\star\|^2, \quad c = \frac{\sigma_{\min}(A)^2}{\|A\|_F^2}.$$

That’s only a statement about *one* step, conditional on where we happen to be. To turn it into an unconditional guarantee after k steps, apply the law of total expectation repeatedly:

$$\begin{aligned} E[\|x_2 - x^\star\|^2] &= E\left[E[\|x_2 - x^\star\|^2 | x_1]\right] \leq E[(1 - c)\|x_1 - x^\star\|^2] \\ &= (1 - c) E[\|x_1 - x^\star\|^2] \leq (1 - c)^2 \|x_0 - x^\star\|^2. \end{aligned}$$

Repeating this argument (tower property, one step at a time) gives, for every k ,

$$E[\|x_k - x^*\|^2] \leq (1 - c)^k \|x_0 - x^*\|^2,$$

— exactly the compounded bound we *plotted* before. Now we know why “applying the one-step bound repeatedly” is legitimate: it’s the law of total expectation, used k times.

```
import numpy as np

rng = np.random.default_rng(151)
m, n = 20, 8
A = rng.standard_normal((m, n))
x_star = rng.standard_normal(n)
b = A @ x_star
row_norm_sq = (A ** 2).sum(axis=1)
p_row_norm = row_norm_sq / row_norm_sq.sum()
sigma_min = np.linalg.svd(A, compute_uv=False)[-1]
frob_sq = row_norm_sq.sum()
c = sigma_min**2 / frob_sq

num_steps = 60
num_runs = 2000
checkpoints = [0, 10, 20, 30, 40, 50, 60]
errors_sq = np.zeros((num_runs, num_steps + 1))

for r in range(num_runs):
    x = np.zeros(n)
    errors_sq[r, 0] = np.linalg.norm(x - x_star) ** 2
    idx_sequence = rng.choice(m, size=num_steps, p=p_row_norm)
    for t, i in enumerate(idx_sequence, start=1):
        a_i, b_i = A[i], b[i]
        x = x + (b_i - a_i @ x) / row_norm_sq[i] * a_i
        errors_sq[r, t] = np.linalg.norm(x - x_star) ** 2

mean_error_sq = errors_sq.mean(axis=0)
initial_error_sq = errors_sq[:, 0].mean()

print(f"{'k':>4} {'empirical E[||x_k-x*||^2]':>28} {'bound (1-c)^k ||x_0-x*||^2':>28}")
for k in checkpoints:
    bound = initial_error_sq * (1 - c) ** k
    print(f"{'k':>4} {'mean_error_sq[k]:>28.5f} {'bound:>28.5f}")
```

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k	empirical E[x_k-x* ^2]	bound (1-c)^k x_0-x* ^2
0	14.24286	14.24286
10	5.34193	9.61687
20	2.26812	6.49338
30	1.00931	4.38437
40	0.45129	2.96036
50	0.20803	1.99885
60	0.09983	1.34964

13 Independence

To define independence for random variables, we build on our previous definition for events.

i Definition: independence of a random variable from an event

X is independent of event A if

$$p_{X|A}(x) = p_X(x) \quad \text{for every } x,$$

i.e. learning that A occurred tells you nothing new about X . Equivalently, $P(X = x, A) = P(X = x)P(A)$ for every x — the L2 definition of independent events, applied to $\{X = x\}$ and A .

14 Independence of two random variables

i Definition: independence of random variables

X and Y are **independent** if

$$p_{X,Y}(x, y) = p_X(x) p_Y(y) \quad \text{for all } x, y,$$

equivalently $p_{X|Y}(x | y) = p_X(x)$ whenever $p_Y(y) > 0$ — conditioning on Y never changes X 's distribution.

Revisiting our RK example: Recall randomized Kaczmarz samples a row index $I_k \in \{1, \dots, m\}$ at every iteration from a fixed PMF p_I (e.g. the row-norm PMF seen previously), **with replacement** — each iteration's draw doesn't remove that row from future consideration.

Because the draws are made independently at each iteration, the joint PMF of the indices at two iterations factors:

$$p_{I_1, I_2}(i, j) = p_I(i) p_I(j).$$

Contrast this with *without-replacement* row-sampling — there, removing a row after sampling it means the joint PMF of two draws does **not** factor this way (the second draw's distribution depends on what the first draw was).

15 Conditional Independence

i Definition: conditional independence

X and Y are **conditionally independent given event C** if

$$p_{X,Y|C}(x, y) = p_{X|C}(x) p_{Y|C}(y) \quad \text{for all } x, y.$$

⚠ Recall: these are genuinely different notions!

A pair of random variables can be independent but not conditionally independent given some C , or conditionally independent given C without being (unconditionally) independent — exactly like the two-coin examples from our previous independence discussion.

16 A key fact: $E[XY] = E[X]E[Y]$ for independent X, Y

i Fact

If X, Y are independent, $E[XY] = E[X]E[Y]$.

Proof: apply two-variable LOTUS with $g(x, y) = xy$, then use independence:

$$\begin{aligned} E[XY] &= \sum_{x,y} xy p_{X,Y}(x, y) = \sum_{x,y} xy p_X(x) p_Y(y) = \left(\sum_x x p_X(x) \right) \left(\sum_y y p_Y(y) \right) \\ &= E[X]E[Y]. \quad \blacksquare \end{aligned}$$

! Independence is essential here

Contrast with linearity of expectation, which needed nothing. In general $E[XY] \neq E[X]E[Y]$: e.g. take $Y = X$, so $E[XY] = E[X^2] \neq (E[X])^2 = E[X]E[Y]$ unless $\text{Var}(X) = 0$.

17 Variance of a sum (with proof!)

i Theorem (stated previously): for independent X, Y ,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

Proof: let $\mu_X = E[X], \mu_Y = E[Y]$. Expanding and using linearity twice,

$$\text{Var}(X + Y) = E[((X - \mu_X) + (Y - \mu_Y))^2] = \text{Var}(X) + \text{Var}(Y) + 2E[(X - \mu_X)(Y - \mu_Y)].$$

Expanding the cross term and applying the fact above,

$$E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - \mu_X \mu_Y = E[X]E[Y] - E[X]E[Y] = 0,$$

so $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

! The cross term

That cross term, $E[(X - \mu_X)(Y - \mu_Y)]$, is called the **covariance** of X, Y . It vanishes exactly because of independence; in general $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$.

18 Example: mean and variance of the sample mean

Let $X_1, \dots, X_n$ be i.i.d. with mean μ and variance σ^2 , and let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$.

$$E[\bar{X}_n] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \mu \quad (\text{linearity — independence not even needed}).$$

$$\text{Var}(\bar{X}_n) = \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{\sigma^2}{n} \quad (\text{independence used here!})$$

The sample mean is unbiased, and its spread shrinks like $1/n$ — the mathematical reason averaging more samples reduces noise, and the seed of the Law of Large Numbers (coming later this course).

```

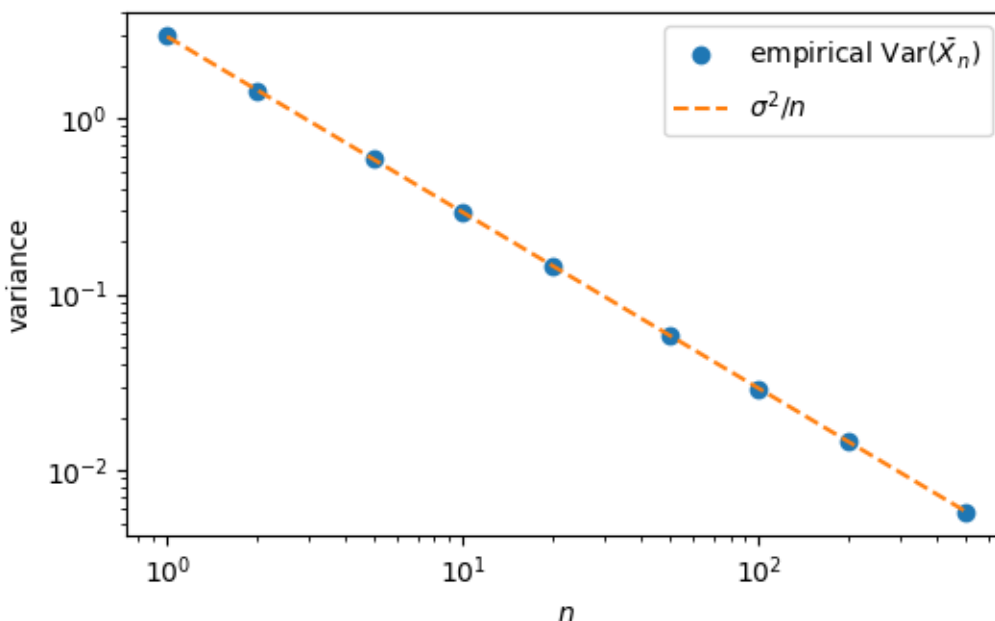
import matplotlib.pyplot as plt

rng = np.random.default_rng(151)
n_values = np.array([1, 2, 5, 10, 20, 50, 100, 200, 500])
repetitions = 20_000
sigma_sq = np.var(np.arange(1, 7)) # variance of a single fair die roll

empirical_var = []
for n in n_values:
    sample_means = rng.integers(1, 7, size=(repetitions, n)).mean(axis=1)
    empirical_var.append(sample_means.var())
empirical_var = np.array(empirical_var)

fig, ax = plt.subplots(figsize=(5.5, 3.5))
ax.loglog(n_values, empirical_var, 'o', label=r'empirical  $\mathrm{Var}(\bar{X}_n)$ ')
ax.loglog(n_values, sigma_sq / n_values, '--', label=r' $\sigma^2/n$ ')
ax.set_xlabel('$n$')
ax.set_ylabel('variance')
ax.legend()
plt.tight_layout()
plt.show()

```



19 Example: unbiasedness of randomized trace estimation

Recall the randomized trace estimator: for $A \in \mathbb{R}^{n \times n}$, draw $z \in \{-1, +1\}^n$ with i.i.d. Rademacher entries and set $\tau = z^\top A z = \sum_{i,j} z_i z_j A_{ij}$. We'll now prove $E[\tau] = \text{tr}(A)$ – this is called **unbiasedness**.

By linearity, $E[\tau] = \sum_{i,j} A_{ij} E[z_i z_j]$. Split into diagonal and off-diagonal terms:

- **Diagonal** ($i = j$): $z_i^2 = 1$ always, so $E[z_i^2] = 1$.
- **Off-diagonal** ($i \neq j$): z_i, z_j independent, so by today's fact, $E[z_i z_j] = E[z_i]E[z_j] = 0 \cdot 0 = 0$

(Rademacher has mean 0).

$$E[\tau] = \sum_i A_{ii} \cdot 1 + \sum_{i \neq j} A_{ij} \cdot 0 = \sum_i A_{ii} = \text{tr}(A).$$